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AI-Directed Prediction of Thermoelastic Wave Propagation in Geological Media

A 2D physics-aware surrogate modelling framework

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Research topic: Physics-aware AI for Thermoelastic Modelling

- 1 Motivation and Problem
- 2 Classical Physical Theory
- 3 Dataset and Surrogate Objective
- 4 AI Models
- 5 Implementation
- 6 Experimental Results
- 7 Conclusion

Why AI surrogates for thermoelastic geological simulations

- 1 Full simulations are expensive.** Coupled transient thermoelastic simulations require fine spatial resolution and repeated parameter sweeps.
- 2 AI surrogates reduce repeated cost.** A trained model approximates field evolution orders of magnitude faster than running the numerical solver each time.
- 3 Physics remains the reference.** The goal is not to replace physics, but to use it for problem formulation, validation, and interpretation.

Question

Can neural surrogate models reproduce the main physical patterns of *thermoelastic wave propagation in 2D plane-strain geological media*?

1. Input

Geometry
Material params
Source conditions

2. Physics

Heat conduction
Thermal expansion
Elastic motion

3. Surrogate

PINN
FNO
MeshGraphNet
Transformer

4. Output

Temperature T
Displacement
(u, v)
Directional response

A planar geological domain $\Omega \subset \mathbb{R}^2$, fixed at $1 \text{ m} \times 1 \text{ m}$, exposed to a localised thermal source.

- **Domain.** 2D plane-strain rock cross-section.
- **Excitation.** Point thermal source at temperature $T_{\text{src}} = 1500 \text{ K}$, reference state $T_{\text{ref}} = 273.15 \text{ K}$.
- **Response.** Coupled temperature T and in-plane displacement (u, v) at a chosen probe.
- **Task.** Estimate the probe response without rerunning the full COMSOL solver.

Locked operating envelope

Source type **point**, mechanical **free**, boundary **prototype simplified**. Trained on 2D plane-strain only; cannot be pushed out of distribution from the client side.

- 1 **Thermal excitation.** A local source changes the temperature field.
- 2 **Thermal expansion.** Temperature change produces local material expansion.
- 3 **Stress formation.** Constrained expansion creates stress gradients.
- 4 **Elastic motion.** Stress gradients accelerate the displacement field.
- 5 **Wave propagation.** The response travels through the geological medium.

This chain is the same in classical thermoelasticity and in our surrogate models; the difference is only *how* each stage is represented.

Section 1 – Classical Physical Theory

Governing equation

$$k \nabla^2 T - C \frac{\partial T}{\partial t} = \gamma T_0 \frac{\partial \varepsilon_{kk}}{\partial t}$$

T – temperature field; k – thermal conductivity; C – volumetric heat capacity; γ – thermoelastic coupling; T_0 – reference temperature; ε_{kk} – volumetric strain.

Heat diffuses from the heated region toward colder zones, and mechanical deformation of the rock additionally drives temperature variation. This coupling is essential for modelling thermoelastic waves in geological media.

Constitutive law (Duhamel–Neumann)

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij} - \gamma \delta_{ij} (T - T_0)$$

Stress tensor. Internal forces produced inside the material under deformation or heating.

Elastic part. First two terms: linear elastic response from volumetric strain ε_{kk} and shear ε_{ij} .

Thermal part. Last term: thermal expansion. Constrained expansion creates thermal stress.

σ_{ij} stress, ε_{ij} strain, δ_{ij} Kronecker, λ, μ Lamé parameters, $\gamma = 3K\alpha$ coupling coefficient.

Dynamic balance in an elastic medium

$$\rho \frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i$$

Substituting the Duhamel–Neumann stress relation connects T , ε , σ , and (u, v) in a single coupled thermoelastic system.

$$T \longrightarrow \Delta T \longrightarrow \varepsilon \longrightarrow \sigma \longrightarrow u$$

Symbol	Meaning	Role
T	Temperature	Thermal state of the medium
(u, v)	In-plane displacement	Mechanical motion (2D)
σ_{ij}	Stress tensor	Internal force response
ε_{ij}	Strain tensor	Local deformation
ρ, C_p, k	Density, heat capacity, conductivity	Thermal material properties
α, λ, μ	Expansion, Lamé parameters	Thermoelastic material properties
γ	$3K\alpha$ coupling coefficient	Thermomechanical coupling
Q, f	Heat source, body force	Excitation terms

All quantities live in a planar 2D plane-strain domain $\Omega \subset \mathbb{R}^2$ in the current prototype.

COMSOL-derived ground truth

Numerical 2D plane-strain simulations provide the training set: geometry, mesh, material properties, and time-dependent thermal / displacement fields exported per node.

- **Inputs.** Geometry, mesh, material properties (granite, sandstone, limestone, basalt), source position, observation time.
- **Outputs.** Temperature T , displacement components (u, v) , stress / strain diagnostics.
- **Pilot dataset.** Sandstone rod-in-collar experiment (4448 nodes, $\Delta t = 10^{-4}$ s, 101 time steps); extended to per-material 2D batches.

Replace repeated COMSOL runs with a fast field prediction.

- 1 Reference simulation.** COMSOL produces time-dependent physical fields under the locked thermoelastic operating envelope.
- 2 Model training.** Each surrogate learns from physical parameters and exported fields; PINN additionally enforces PDE residuals.
- 3 Fast inference.** New scenarios are evaluated through a unified REST contract without rerunning the full solver.

Section 2 – AI Models

Core idea

- Point-wise MLP
 $f_{\theta}(x, y, t, \text{material}) \rightarrow (T, u, v)$.
- Composite loss: supervised data MSE, velocity-consistency, elastic-wave residual, coupled heat-equation residual.
- Penalises physically inconsistent outputs even when data coverage is limited.

Why it fits

- Governing equations are known.
- Physical consistency is a key validation criterion.
- Works as the theory-aware baseline.

Limit

Training is slow; residual weighting is delicate.

Core idea

- Operator-learning model for field-to-field prediction.
- Captures global spatial patterns through spectral convolutions.
- Efficient on structured-grid PDE data.

Conceptual workflow: input field → frequency-space patterns → learned operator → predicted field.

Current caveat

2D-adapted branch. Output normalisation is not yet calibrated: absolute temperature and displacement scales are outside the physical band. Direction sign also needs verification. *Diagnostic branch, not a physical reference yet.*

Graph-based surrogate for irregular meshes

- 1 Geometry as a graph.** Mesh nodes become graph nodes; mesh connectivity becomes graph edges.
- 2 Local interaction.** Messages propagate information between neighbouring regions of the domain.
- 3 Fits geology.** Irregular rock geometry and heterogeneous media are naturally represented by graphs.

Limit

Current implementation is a fixed-horizon rollout; the requested observation time does not yet drive the number of message-passing steps. Absolute magnitudes inherit a 2D-dataset / 3D-checkpoint padding mismatch.

How it can be used

- OFormer-style encoder-decoder with cross-attention.
- Tokens carry coordinates, current-state fields, and material properties.
- Predicts (T, u, v) at queried coordinates.

Strengths

Flexible coordinate-query model; the fastest route at inference time in the latency test.

Limit

Current loss is supervised MSE only; no physics-informed residuals yet. High memory demand for dense fields.

Model	Physics	Accuracy	Speed	Generalisation	Interpre
PINN	High	Medium–High	Medium	Medium	Hi
FNO	Medium	Pending calibration	High	High on similar grids	Med
MeshGraphNet	Medium	Medium–High	Medium–High	High on irregular meshes	Med
Transformer	Low–Medium	High with data	High	Medium	Low–M

Framing: PINN as the physics-consistency baseline, MeshGraphNet for mesh-aware prediction, Transformer for fast flexible queries, FNO once normalisation is corrected.

Section 3 – Implementation

Service-oriented prototype (docker-compose)

Service	Port	Role
frontend	8080	Vanilla HTML/CSS/JS single-page app, nginx
backend	8000	FastAPI orchestrator, unified v2 contract
pinn-service	9003	PINN MLP, supervised + physics residuals
fno-service	9002	FNO2d, supervised MSE
mgn-service	9001	MeshGraphNet rollout
transformer-service	9004	OFormer-based autoregressive operator

Each model service exposes the same prediction endpoint; the backend validates, resolves the geological medium from the catalogue, routes, and normalises the response.

Request shape

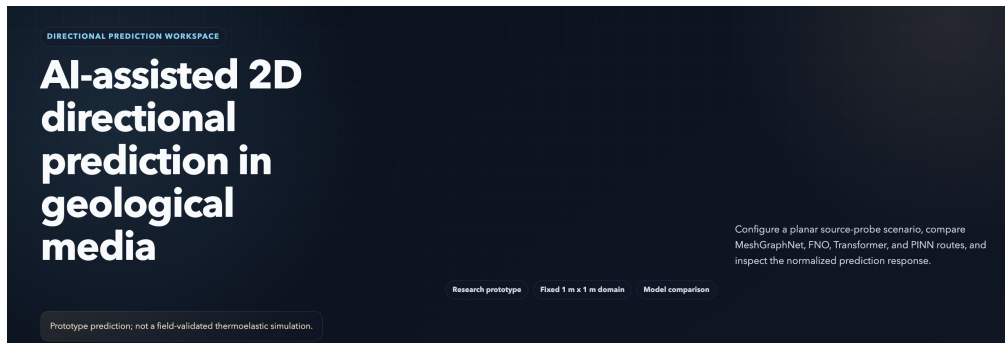
Schema-versioned payload with (x, y) source and probe inside the fixed $1\text{ m} \times 1\text{ m}$ unit square; observation time in seconds; locked scenario block (point thermal source, free mechanical constraint, prototype boundary).

Response shape

Thermal (T, θ) , in-plane displacement (u, v) with magnitude, directional response (distance, azimuth, unit direction), temporal (travel time), model metadata, diagnostics.

Locked invariants

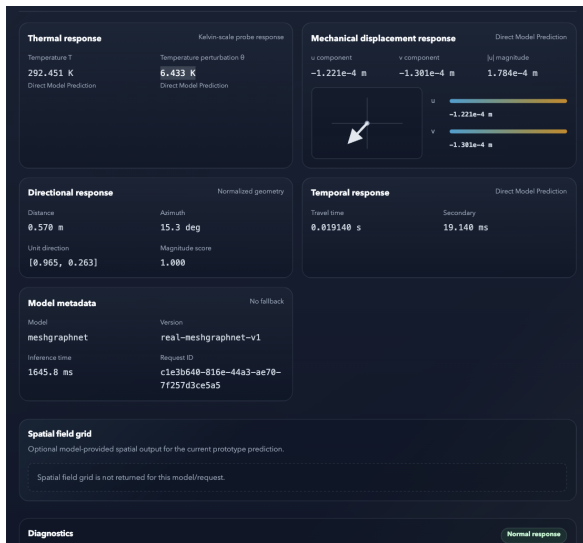
$T_{\text{ref}} = 273.15\text{ K}$, $T_{\text{src}} = 1500\text{ K}$, domain $1\text{ m} \times 1\text{ m}$, no source frequency. Resolved server-side so clients cannot push trained models out of distribution.



Static single-page application; no runtime framework. Three contextual chips pin the scope: *research prototype*, *fixed $1\text{ m} \times 1\text{ m}$ domain*, *model comparison*. Every response carries a prototype-prediction disclaimer.



- Source (warm disc) and probe (cool disc) inside the unit square.
- Propagation arrow carries distance and azimuth.
- Unit-direction key in the bottom-left.
- Geometry computation is the same one the backend uses, so the SVG and numeric panels never disagree.



Cards report the thermal response at the probe (T and perturbation $\theta = T - T_0$), the in-plane displacement (u, v) with magnitude, the directional response, the temporal response (travel time), and the model metadata. The diagnostics strip carries the run state, schema version, and any warnings returned by the backend.

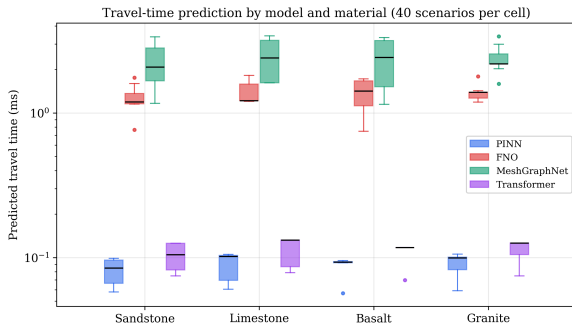
Section 4 – Experimental Results

Evaluation setup

- 1 **40 scenarios per material**, four materials (granite, sandstone, limestone, basalt).
- 2 **$40 \times 4 \times 4 = 640$ model responses** collected under the same v2 contract.
- 3 **Zero fallback**: every plotted response is an active model output, not a mock.
- 4 **2D plane-strain only** – the operating envelope of all four trained surrogates.

FNO caveat

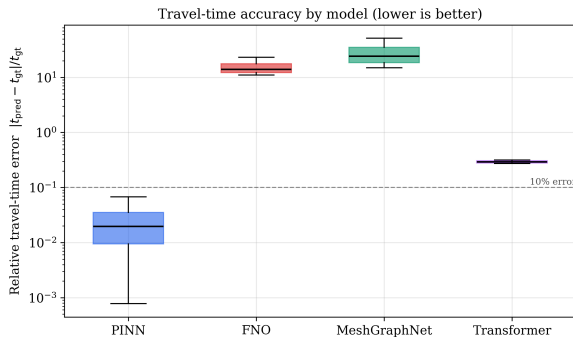
FNO is loaded, but its absolute temperature and displacement scales are outliers. Treated as a diagnostic branch in this comparison.



Mean travel time per route:

- PINN ≈ 0.11 ms
- Transformer ≈ 0.14 ms
- FNO ≈ 1.29 ms
- MeshGraphNet ≈ 2.47 ms

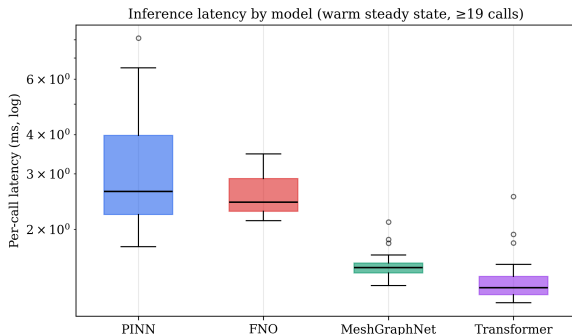
Sandstone is slightly slower than basalt for every route, which is consistent with its lower V_p . Model-family differences dominate the spread over material differences.



Reference: analytical body-wave travel time r/V_p .

Model	Median rel. error	Within 10 %
PINN	2.0%	100%
Transformer	29.2%	0%
FNO	1403.6%	0%
MGN	> 100%	0%

PINN is the only route within the 10% band on every scenario of the balanced sweep.

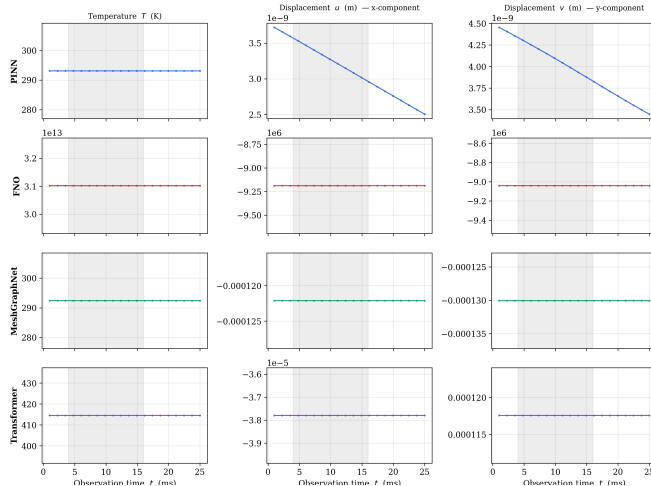


All four routes return in under 10 ms on the development machine.

- Transformer ≈ 1.3 ms (median)
- MeshGraphNet ≈ 1.5 ms
- FNO ≈ 2.4 ms
- PINN ≈ 2.6 ms (longer upper tail)

Latency is reported as a prototype-level wall-clock measurement, not a production benchmark.

Model predictions vs observation time — Sandstone
source = (0.20, 0.50) m, probe = (0.80, 0.50) m · shaded region = training time range (4-16 ms)



PINN is the only route that changes smoothly with the requested observation time – t is a continuous input coordinate of the model.

FNO, MeshGraphNet, Transformer behave as fixed-horizon operators in the current implementation: the probe value remains almost unchanged across the tested time range.

This is an architectural consequence, not a plotting artefact.

- 1 **Limited training data.** The COMSOL-derived pilot configuration is not a full field-validated geological database; generalisation beyond the trained envelope is not yet established.
- 2 **FNO normalisation.** The current 2D adaptation has inconsistent output scales for several metrics; not a physically reliable route yet.
- 3 **Prototype-level latency.** Numbers reflect the development machine, not a production deployment.
- 4 **Analytical reference is isotropic.** r/V_p is useful as a sanity check but is not a substitute for a full thermoelastic validation study.

Section 5 – Conclusion

What works

All four services respond successfully; no fallback responses are used. The v2 contract supports a fair end-to-end comparison across model families.

Model behaviour

PINN: physically stable, only route inside the 10% accuracy band. MGN: stable but fixed-horizon. Transformer: fast and material-sensitive. FNO: integration branch only, pending calibration.

Material effect

Sandstone is generally slower than basalt across routes. Model-family differences still dominate the spread over material differences.

- 1 **Use PINN as the physics-informed baseline** for the directional prediction problem in this thesis.
- 2 **Use MeshGraphNet** as the mesh-aware surrogate candidate.
- 3 **Use the Transformer route** as the fast coordinate-query candidate.
- 4 **Correct FNO output normalisation** and re-run the balanced basalt/sandstone/granite/limestone sweep on the same pipeline.
- 5 **Add physics-informed residuals** to the Transformer route and time-conditioned rollout to MeshGraphNet.

Physics-aware surrogates,
faster geological prediction.

Thank you for your attention

Questions?





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